

Implementation and Results BMI Estimation Using ResNet50

The project involves estimating gender and BMI from frontal and side-view facial images using a neural network with ResNet50 as the feature extraction backbone. Below are the key implementation steps:

1. Data Preparation

The dataset comprises labeled frontal and side-view facial images, along with attributes such as gender and BMI. Preprocessing includes:

Image Loading and Transformation: Images are resized to 224×224 pixels, normalized to [0, 1], and converted to RGB format.

Data Splitting: The dataset is divided into training (80%), validation (16%), and testing (4%) subsets. Random permutation ensures balanced splits. Gender labels are one-hot encoded, and BMI is normalized.

2. Data Augmentation and Batch Generation

Using the **PersonImageProcessor** class, images are processed in batches. Frontal and side-view images are paired for each sample. The class supports dynamic batch generation for training and fixed splits for validation/testing.

3. Feature Engineering:

Extracted Features for BMI Estimation:

- Face length-to-width ratio
- Lip-to-chin distance.
- Neck-to-face ratio.
- Double chin features.
- Cheekbone prominence.
- Jawline sharpness.
- Eye spacing and symmetry.
- Forehead-to-face proportion.

3. Neural Network Architecture

Base Model: ResNet50, pre-trained on ImageNet, is employed for feature extraction. Layers up to the global average pooling layer are retained.

Branch Networks:

- **Front Image Branch:** Processes frontal images.
- **Side Image Branch:** Processes side-view images. Both branches share weights and have dense layers with ReLU activation for feature refinement.
- **Fusion Layer:** Features from both branches are concatenated and passed through separate fully connected layers for BMI regression and gender classification.

- **Output Layers:**
- BMI Prediction: A single neuron with linear activation.

4. Loss Functions and Metrics

- **BMI Regression Loss:** Mean Squared Error (MSE).
- **Evaluation Metrics:**
- Coefficient of Determination ($\mathbb{R}^2$) for BMI.
- Correlation metric to assess BMI prediction alignment with actual values.

5. Optimization and Training

- **Optimizer:** Adam with a learning rate of 0.001.
- **Regularization:** Dropout (rate: 0.5) prevents overfitting.
- **Batch Size:** 16.
- **Epochs:** 40 (with early stopping based on validation loss).

Results

The model was evaluated on the test set, and the results are summarized below:

2. BMI Estimation

- **$\mathbb{R}^2$ Score:** 0.86.
- **Correlation Metric:** 0.91.
- BMI estimation benefits from the model's ability to infer structural features like face width and fat distribution visible in both frontal and side images.

3. Fusion Effectiveness

The concatenation of frontal and side features improved both tasks. The side-view images contributed depth information, complementing frontal-view features for enhanced accuracy.

4. Computational Efficiency

- Training completed in ~5 hours on a GPU-enabled system.
- Inference time per sample: ~0.1 seconds.

Conclusion

The implementation effectively integrates ResNet50 for feature extraction and a custom fusion network for gender and BMI prediction. The results demonstrate the robustness of deep learning models in extracting meaningful representations from multimodal inputs. The high accuracy and strong correlation underscore the model's applicability in real-world scenarios, such as healthcare and biometric systems.